

Plant calendar — Detailed design

006_DD version 1 implements SCR-006, FN-037–040 and REQ-070–075 under WI-010 approved plan revision 1 step 6. Main DD review package only; the DD family and implementation authorization remain incomplete.

1. Document control

Field	Value
Document ID / version	006_DD / 1
Category / system / subsystem	UI and module design / ProductionManagementAI / Master data
Screen	SCR-006 「稼働カレンダー」 (Working calendar)
Author / work item	Agent / WI-010
Created / updated	2026-10-02 / 2026-10-02
Inputs	Approved 006_REQ, 006_BD and 006_DB version 1
Review state	Submitted for review; not implemented

Version	Date	Author	Change
1	2026-10-02	Agent	Initial fields, modules, state transitions and visual companions

2. References and design ownership

Reference	Purpose
006_REQ	REQ-070–075 and accepted RP-01–05
006_BD	Routes, navigation and calendar regions 1–9
006_DB	Three-table storage, exact checks, snapshots, locks and recovery
WI-010_plan	Sequential review and phase authorization
Layering_ADR	Layered controller/use-case/persistence structure

Reference	Purpose
Authentication ADR	Existing same-origin cookie/roles

Companion design documents

All three companions are mandatory and reserved below, without broken links to nonexistent files. Each is created only after the preceding review. Referenced by: none yet; later companions will reference this immutable approved main DD.

Reserved file	Owns	Plan step
006_DD-API_稼働カレンダー.md	Endpoint field catalogs, bounds, errors, roles and endpoint telemetry	7 after main DD approval
006_DD-FN_稼働カレンダー.md	Backend methods, resolver, exact transactions and budgets/instrumentation	8 after API approval
006_DD-SPD_稼働カレンダー.md	Client processing sequences, detailed focus/keyboard flows and final state mockups	9 after FN approval

Function	Main DD ownership	Later specification
FN-037	Applied month/scope/day/read state	API read contract, FN snapshot, SPD sequencing
FN-038	Weekly fields and current/future/past variant	API payload, FN version transition, SPD editor flow
FN-039	Exception fields, remove dialog and retained draft	API target/errors, FN atomic marker append, SPD confirmation
FN-040	Capacity selection and honest result state	API transport, FN decimal floor/eligibility, SPD stale-result guard

3. Component organization and module design

X-1. Paths

Planned path / namespace	Responsibility
src/frontend/src/features/plant-calendar/	Calendar route, editors, state, API adapter and validators
src/frontend/src/features/production-orders/messages.ts	Existing single Japanese catalog additions
src/frontend/src/components/icons.ts	Central CalendarRange alias proposed, distinct from Factory/Package/ClipboardList; verify installed export later
src/backend/ProductionManagementAI.Domain/PlantCalendar/	Calendar snapshots and row-local invariants
src/backend/ProductionManagementAI.Application/PlantCalendar/	Use cases/ports, methods specified by DD-FN

Planned path / namespace	Responsibility
src/backend/ProductionManagementAI.Infrastructure/PlantCalendar/	EF mappings, snapshot reader, aggregate writer
src/backend/ProductionManagementAI.Api/Controllers/	PlantCalendarController; exact contracts in DD-API

X-2. Shared components

Reuse AppHeader, AppNavbar, GuardedLink, NavigationGuardProvider, same-origin apiClient and existing session behavior. Dates stay date-only; existing Japanese number formatting is display-only, never numeric validation. No new component kit, provider, state library or i18n package.

X-3. Feature modules

These are proposed names, not existing code. All authored by Agent on 2026-10-02.

Module / inputs	Preconditions	Responsibility / return	Dependencies
PlantCalendarPage / applied URL key	Admin/Operator session	React view; month table/phone agenda, selected-day and capacity context	Calendar API adapter, header, catalog
WeeklyPatternEditor / authoritative definition+revision	Read succeeded; server date known	React form; independent draft, weekdays/start, retained version inspection	Validation, navigation guard
DateExceptionEditor / scope/date/target+revision	Exact captured identity; editable server rules	React form; state/hours/reason, remove intent, preserved draft	Validation, confirmation
CapacityPanel / line/product/date query	Today/future; current reference choices	React view; request-keyed result or explicit unavailable	Calendar API adapter, formatters
CalendarConfirmation / captured action+invoker	Remove/withdraw/discard intent	Centered native dialog; Cancel first, focus return, no write on dismissal	Native dialog and route guard
CalendarDraftValidation / mode+strings+context	Typed input and observed date; no authoritative claim	Structured field errors / valid normalized draft	Exact decimal/date rules and catalog
PlantCalendarController / request	Existing cookie and Admin/Operator on every read/write	HTTP result, no domain SQL/algorithm in controller	Application port, DD-API catalog

X-4. Rule details

CalendarDraftValidation is a screen-owned validator, not an XML rule engine. Rule type: synchronous typed validation, not authoritative concurrency checking. Rule references: HoursTextRule, DateTextRule, ReasonTextRule and the server's editable context. Condition references: observed plantToday/timezone/activation, selected

line state, captured target and aggregate revision; these observations may become stale, so the server always rechecks. Check parameters are F-04–11 and capacity F-14. Inputs are exact text/booleans/weekday set; output is a field error map or normalized draft. It never writes, increments revisions or merges.

X-5. Includes / legacy concepts

ASPX include paths, XML containers, condition files and server-rendered HTML includes: not applicable to React/.NET. Backend business methods belong to FN; step-by-step UI sequences belong to SPD, avoiding duplicate algorithm ownership.

4. Screen layout, regions and mockups

Reuse approved BD month/phone regions 1–9. The following editor SVGs add precise field ownership; numbers 10–16 complement BD, while 2 retains the message area. The desktop sheet combines alternative editor modes: regions 11/15 are weekly-only, 12/13 are date-only; the application shows one mode at a time.

稼働カレンダー — 編集領域

2

状態・メッセージ

10

対象 : L-001 / 日付 : 2026-10-12

11

週間規則 : 月・火・水・木・金

12

日付例外 : 稼働 / 4 時間

13

理由 : 保守後の追加稼働

14

保存 / キャンセル / 削除

15

保持版・予約規則 (過去は参照のみ)

16

確認 : キャンセルを初期フォーカス

稼働カレンダー — 編集領域

2

入力内容を確認してください

10

L-001 / 2026-10-12

12

稼働 / ライン設定を使用 / 時間

13

理由（任意）

14

保存 / キャンセル / 削除

16

確認：対象と操作を明記

Region	Fields / content	Notes
2	F-16 status/errors	Announced status or error summary; never assert uncertain commit

Region	Fields / content	Notes
10	F-04/F-07 effective date or captured scope/date	Weekly start editable for new date; exception identity immutable on edit
11	F-05 weekday choices	Seven labeled booleans; all closed valid
12	F-08–10 working state/hours	Explicit or line-inherited; closed has no stored hours
13	F-11 reason	Optional plain text, 500 code points
14	F-12–13 actions	Save/Cancel/remove or future withdrawal; draft/permission/pending guards
15	F-06 retained/current/future definitions	Read-only past payloads, current selection, future correction/withdrawal
16	Confirmation	Captured target/version; safe initial focus and centered viewport

Mockup artifact: [English-caption static states](#) and [Japanese-caption static states](#). Nine artboards: unconfigured read, populated calendar/capacity, weekly edit, exception validation, past read-only, confirmed success, conflict, unknown outcome and confirmation. UI text is Japanese in both editions. Layout samples use Oct 12 L-001 reopening to 4 hours, P-1001 at 0.125 min/個 = 1,920 個. These are visual design samples; no seed, server, real execution or physical-phone claim. Static spans illustrate controls, not a functional form. Browser render and phone-width inspection validate the artifact only. Final SPD owns richer per-state processing/focus mockups. No external artifact publishing required.

5. Fields, normalization and validation

BD field	Stored/display input	Initial / validation / behavior
F-01 month	YYYY-MM text	URL or server month; impossible month rejected; applied value separate from draft
F-02 scope	Plant/null or UUID line	Validate reference; active choices for new use; retained retired scope readable
F-03 selected day	YYYY-MM-DD + resolved state/source	Bound to applied month; before coverage unavailable; no synthetic days
F-04 weekly start	YYYY-MM-DD text	New default server today; past immutable; future correction keeps identity date
F-05 weekdays	Seven boolean choices	Existing mask or Mon–Fri initial; all false valid; server maps distinct set to 0–127
F-06 definitions	Current/retained/future payloads	No editing old revisions; withdrawal future only; first activation protected
F-07 exception identity	scope/date and optional target revision id	Existing identity immutable; move means separate new exception
F-08 state	Working/closed	Explicit choice; removal not a user working state
F-09 inherit	Boolean, working only	True means null stored hours and current line baseline; no lower override merge
F-10 hours	Exact decimal string	Required for explicit working; >0, <=24, at most 3 fractional digits; no exponent, grouping, binary float or implicit rounding

BD field	Stored/display input	Initial / validation / behavior
F-11 reason	Unicode text	Trim both ends, blank -> null, <=500 code points; preserve internal content; render as text
F-12 Save/Cancel	Local action	Save only changed valid editable draft; known success returns calendar; dirty Cancel confirms discard
F-13 remove/withdraw	Captured action	Existing exception today/future; future weekly only; centered confirm, version unchanged
F-14 capacity	Line/product/date selections	Current valid choices, today/future; explicit request; changing key invalidates old result
F-15 result	Quantity/unit/hours/time/eligibility	Floor to whole discrete units or 3 decimals kg/m; unavailable not zero; plant-only has no invented hours
F-16 notice/verification	Operation outcome + read observation	Field errors/conflict/unknown keep draft; observation not proof of this client's commit

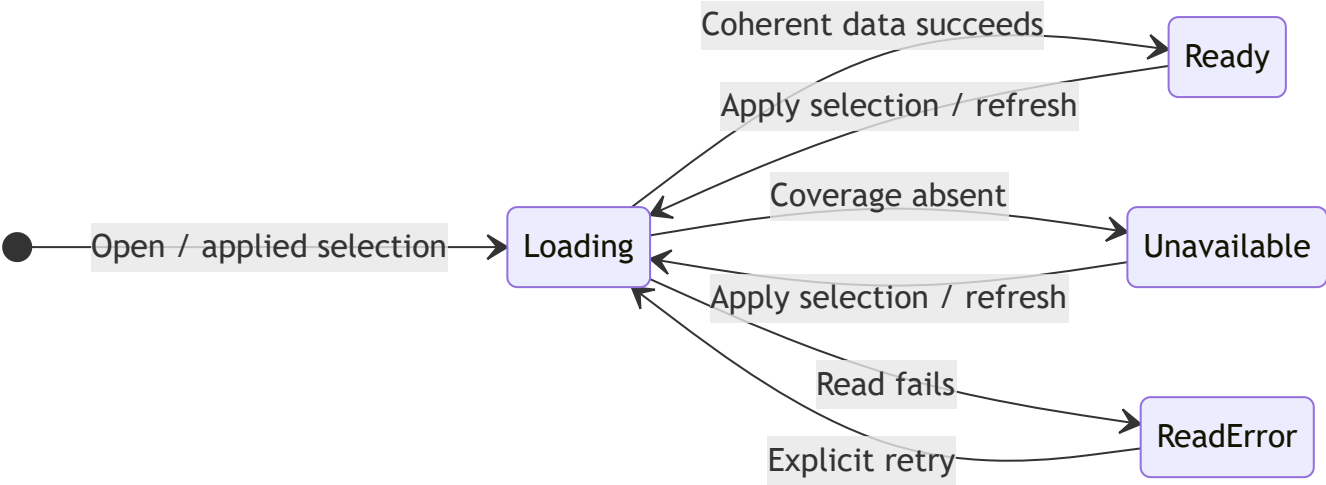
Hours normalization trims surrounding whitespace but does not round or truncate precision. Toggling closed/inherit explicitly clears contradictory hours; hidden values must not be sent. Count reason with Unicode code points, not JS UTF-16 length; reject malformed text under the API contract. Dates are parsed as calendar values, not Date objects interpreted in browser timezone. DB DateOnly bounds are 0001-01-01 through 9999-12-31; exact API query limits remain later ownership.

Every read/write requires Admin/Operator; route visibility is not authorization. Server plantToday and current master state decide editability, including after midnight or retirement. Retired-line exception can be removed today/future, but not newly created/edited. All calendar writes carry observed global revision; independent calendar changes may cause conflict. Do not copy line xmin into it. No draft notes, tokens or inputs in URL/localStorage/sessionStorage.

6. Processing and state transitions

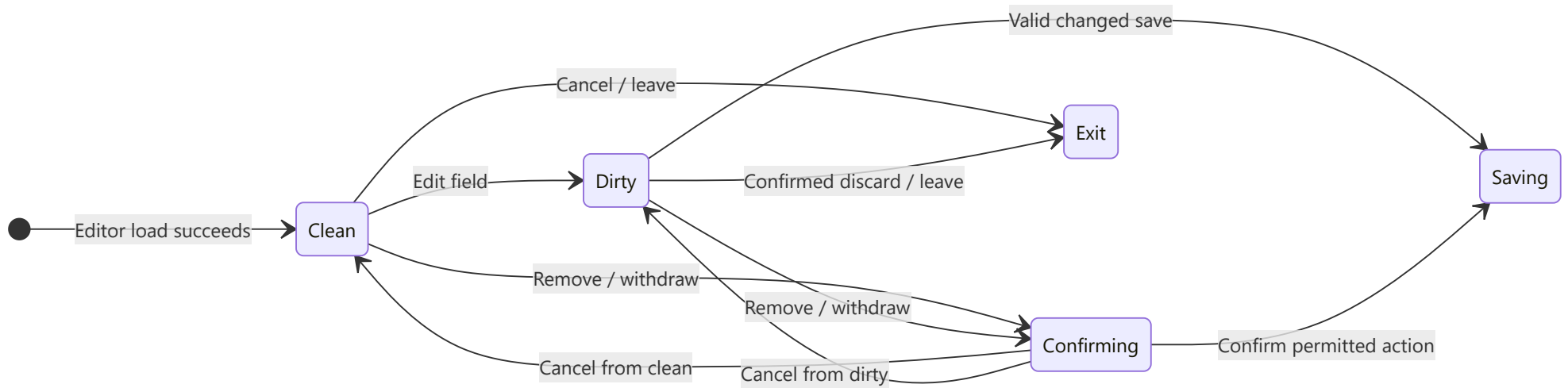
State tokens below are implementation design names. The diagrams show exactly the following edge inventories. Same-state validation/pending/dirty events and any-state auth errors remain in the separate table below. Diagrams split read, draft and write/reconciliation to keep labels readable; shared tokens refer to the same editor state, not separate screen IDs.

Read state — 7 diagram edges



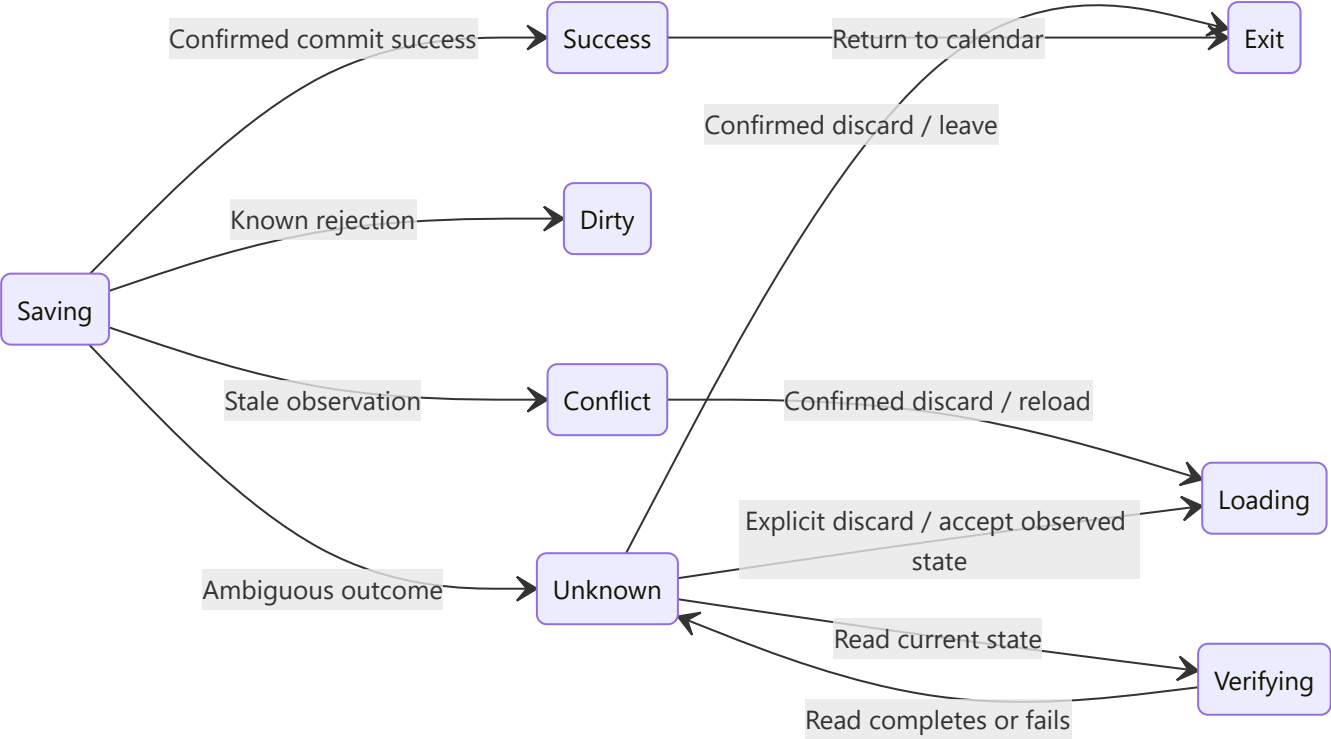
From	Event	To	Effect
[*]	Open / applied selection	Loading	No fabricated data
Loading	Coherent data succeeds	Ready	Bind result to applied key
Loading	Coverage absent	Unavailable	Explicit configuration unavailable
Loading	Read fails	ReadError	Honest error, no relabeled old data
ReadError	Explicit retry	Loading	Read only
Ready	Apply selection / refresh	Loading	Invalidate old result
Unavailable	Apply selection / refresh	Loading	Read only

Editor draft — 10 diagram edges



From	Event	To	Effect
[*]	Editor load succeeds	Clean	Store baseline and revision
Clean	Edit field	Dirty	Local draft only
Clean	Remove / withdraw	Confirming	Capture operation/identity/version
Dirty	Remove / withdraw	Confirming	Capture target; do not silently save draft
Confirming	Cancel from clean	Clean	Restore prior clean state
Confirming	Cancel from dirty	Dirty	Keep draft
Dirty	Valid changed save	Saving	Submit once
Confirming	Confirm permitted action	Saving	Submit captured action once
Clean	Cancel / leave	Exit	No write
Dirty	Confirmed discard / leave	Exit	No write

Write and reconciliation — 10 diagram edges



From	Event	To	Effect
Saving	Confirmed commit success	Success	Refresh calendar after known write
Saving	Known rejection	Dirty	Retain values and field errors
Saving	Stale observation	Conflict	Retain draft; block overwrite
Saving	Ambiguous outcome	Unknown	Block mutation/replay
Conflict	Confirmed discard / reload	Loading	Fetch new authoritative baseline
Unknown	Read current state	Verifying	Read only; retain original draft
Verifying	Read completes or fails	Unknown	Show observation/error; no success inference

From	Event	To	Effect
Unknown	Explicit discard / accept observed state	Loading	Only after verification; no automatic replay
Unknown	Confirmed discard / leave	Exit	Cancel remains available; no write
Success	Return to calendar	Exit	Preserve applied context

From	Event	To / effect
Ready	Select day, valid local capacity selection	Same read state; capacity has independent request-keyed Loading/Ready/Unavailable/ReadError
Clean / Dirty	Field validation or unchanged Save	Same state, no mutation; invalid fields announced/focused
Confirming	Context/version no longer matches	Return prior Clean/Dirty with error; no confirmed operation against different target
Saving / Verifying	Additional write click	Same state; block duplicate mutation
Loading / ReadError	Cancel editor	Exit, no mutation; draft discard if one was retained
Any authenticated state	401 / 403	Existing login / forbidden view; clear feature data and stop processing
Any editable state	Date becomes past / line retires	Show authoritative restriction; retain local input where session allows, no permitted save

Conflict keeps local draft until explicit guarded reload. Unknown verification reads without discarding the draft and always returns to Unknown with an observed state or read error. Accepting observed state requires explicit draft discard; this resets to a new baseline, never asserts the earlier save succeeded and never replays its request. Exact settlement/error handling must be finalized in API/FN. Cancel remains available while Unknown; Saving prevents duplicate actions and accidental in-app destructive departure. Browser exit cannot be guaranteed: use the existing supported beforeunload warning and reconcile lost outcomes.

Flow	Specification owner	Trigger
Applied URL/read cancellation and focus	DD-SPD calendar block	FN-037 Apply/month/day
Draft/save/dirty navigation	DD-SPD editors; DD-FN mutation methods	FN-038/039
Confirmation remove/withdraw	DD-SPD dialog; DD-FN snapshot append	FN-038/039
Unknown verification	DD-SPD result state; DD-API error contract; DD-FN read	F-16
Capacity request/decimal/eligibility	DD-SPD panel; DD-FN capacity	FN-040

7. API use and persistence mapping

This is a logical call inventory, not a premature endpoint field catalog. Proposed base /api/plant-calendar; exact routes/methods are DD-API's next review. No existing production-order/master endpoint change is authorized here.

Logical call	Purpose / ownership
Month/day/context read	Matching scope/dates, activation, current revision and source; DD-API/FN
Weekly versions/history read	Bounded current/future/retained records; DD-API/FN
Weekly save / future withdraw	One observed-revision transition; DD-API/FN
Exception save / remove	One captured scope/date/target transition; DD-API/FN
Line/product choice and capacity read	Valid current references and one dated descriptive result; DD-API/FN
Verification read	Current revision/target/history observations; no replay ledger; DD-API/FN

Operation	Approved DB mapping / transaction	Screen implication
Read/context/capacity	Read-only REPEATABLE READ, projected calendar and current master data	One coherent snapshot, no mixed versions or per-day roundtrips
Weekly mutation	State FOR UPDATE; append/old-current=false; revision+1 atomically	Current-day insertion preserves older starts; future withdrawal marker
Line exception mutation	Line FOR SHARE before state FOR UPDATE	Retirement race rechecked; no reverse lock order
Plant exception mutation	State FOR UPDATE only	Null plant identity uniqueness includes markers
Remove/re-add	Retained snapshot/marker, no DELETE	Fallback restored; past notes/history retained

Prefer clear projected table reads; capacity may join existing line/pair/product within the same snapshot to avoid N+1 and mixed-generation eligibility. Existing query behavior is not rewritten. No stored capacity or order relationship. Column-scoped EF updates must respect DB grants; all service algorithms, query budgets and exact lock/cancellation behavior have one owner in DD-FN.

8. Exceptions, accessibility and observability

Outcome	User behavior	Retry / instrument boundary
Invalid input/reference/past date	Keep input; field/page explanation and first-error focus	Explicit correction only; bounded validation outcome
Known version conflict	Keep draft; guarded reload	No automatic overwrite/merge; concurrency outcome

Outcome	User behavior	Retry / instrument boundary
Read error	Honest retry and unavailable current result	Explicit read retry; operation/outcome/duration
Unknown write outcome	Keep draft; block writes; verify or Cancel	No blind replay, payload-equality success claim or automatic retry
Authentication / authorization	Login or forbidden; feature processing stops	Existing session behavior, no cookie/body logs
Lock timeout/deadlock/known rollback	Retain draft and safe generic error	Bounded transient outcome; exact budgets later

RFC 9457 Problem Details field/error catalog belongs to DD-API. OpenTelemetry endpoint/use-case spans plus count/duration metrics are mandatory; API/FN owns names and bounded operation/outcome labels. Exclude raw dates, line/product IDs, notes, quantity text, revision tokens, SQL internals and credentials from labels/user diagnostics. Runtime exports/timeouts are not configured in this design.

Accessibility: semantic desktop table/native day buttons and mobile ordered agenda; associated form labels, visible focus, text+color states and error/status announcements. Confirmation uses fixed native dialog centered to viewport, forward/reverse Tab trap, Cancel initial focus, Escape and invoker focus restoration. Dirty route/history exits use the existing guard. Layout targets 320 CSS px and 200% zoom; past/retired restrictions have text. Mockup spans do not establish interactive accessibility compliance; later implementation must verify it.

9. Verification viewpoints and next review

Requirement	Meaningful later verification
REQ-070	URL restore, impossible dates, activation unavailable, read cancellation and PC/SP
REQ-071	Today/future replacement, future withdraw/re-add, all closed, protected past/initial definition
REQ-072	Null plant duplicate, exact hours, reason code points, remove/Cancel, retained revisions
REQ-073	Line reopening, full override/hour fallback, current-unit generation, floor precision and unavailable vs zero
REQ-074	Admin/Operator every endpoint, global conflict, midnight/retirement race, unknown outcome, focus/dialog/mobile/zoom and telemetry
REQ-075	Existing order-start, due-date and dashboard-day rules unchanged

Test IDs belong to plan step 11; no unrun tests are reported as passing. No open business blocker. Exact API bounds, backend budgets/algorithms and final SPD processing await their sequential reviews. Migration/recovery remains approved DB; no schema execution in this phase. Submit main DD with its SVGs, mockups and EN/JA PDFs; await approval before writing DD-API. Full family/security gate is pending.